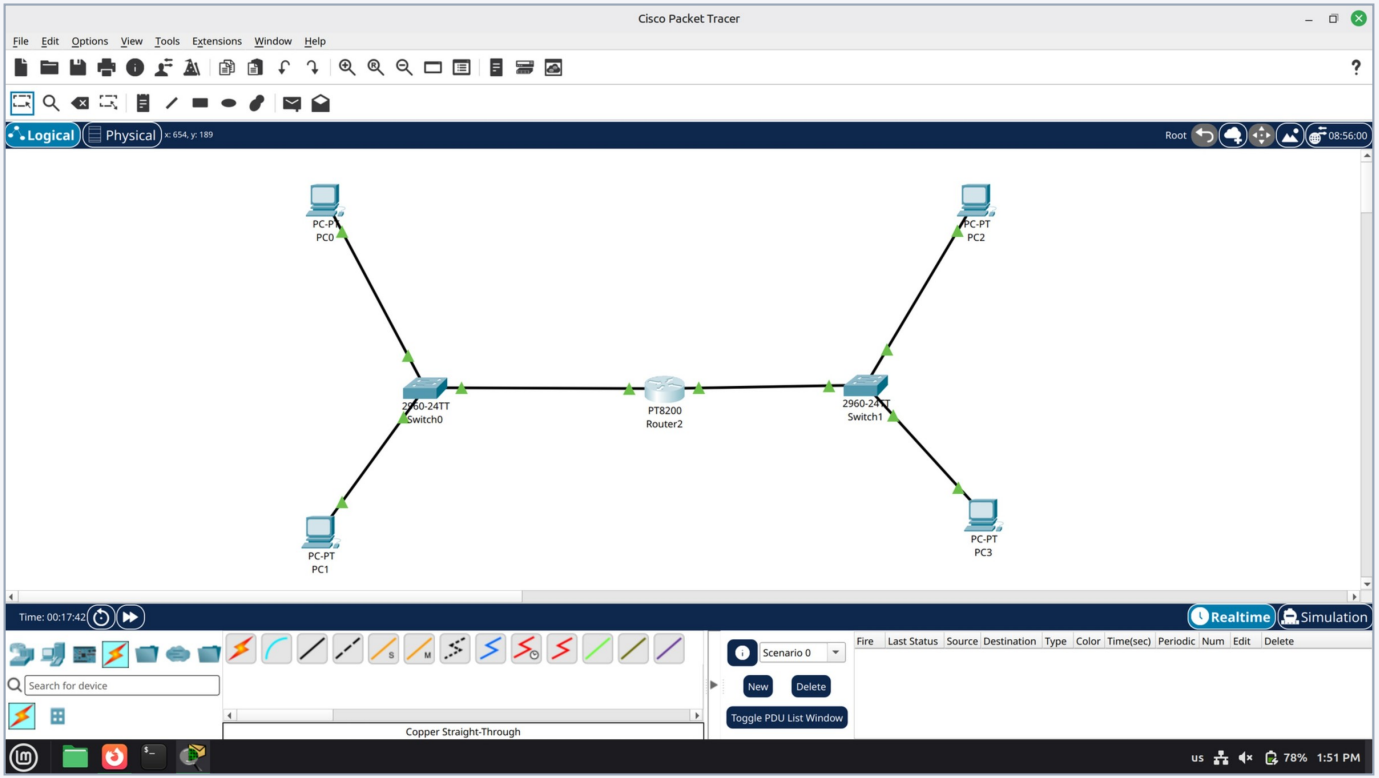
PACKET TRACER LAB · 2 LANS, 1 ROUTER

2nd Task

STUDENT DETAILS	
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Date of submission	Monday, 9 November 2026

TEACHING ASSISTANT
Prof\Toka Farid

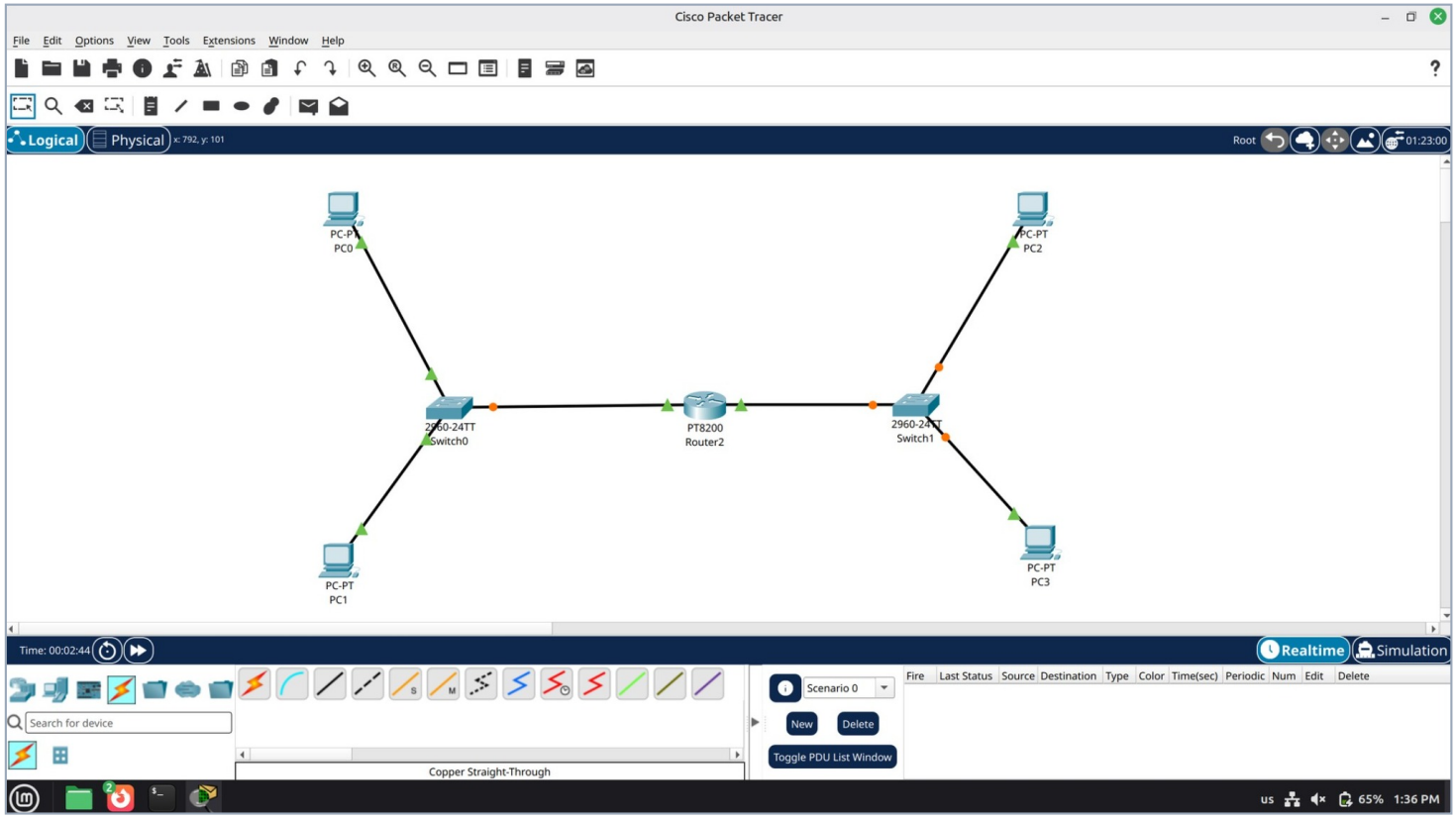
EVALUATION	ACCEPTED	REFUSED
Topology accurate		
Switch checks (Switch0 / Switch1)		
Router configuration		
PC addressing & gateways		
Connectivity tests (ping)		
Documentation & screenshots		



STEP 1 · TOPOLOGY

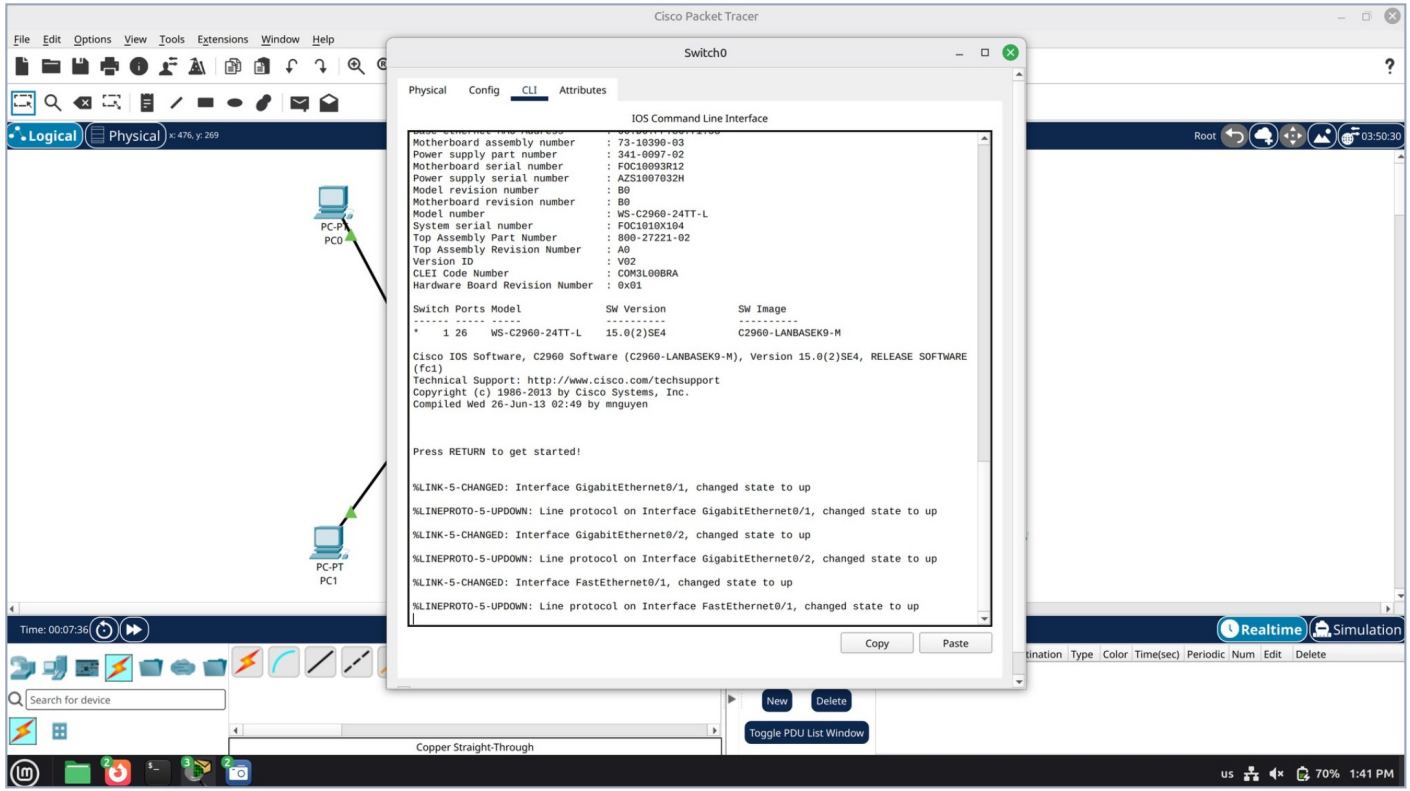
Build the topology

- **Devices:** PC0–PC3, two 2960 switches, one PT8200 router
- **Left LAN:** PC0 and PC1 on Switch0
- **Right LAN:** PC2 and PC3 on Switch1
- **Cables:** Copper Straight-Through, with Router2 between the switches
- Orange dots are ports still converging; wait for green



Check switch links

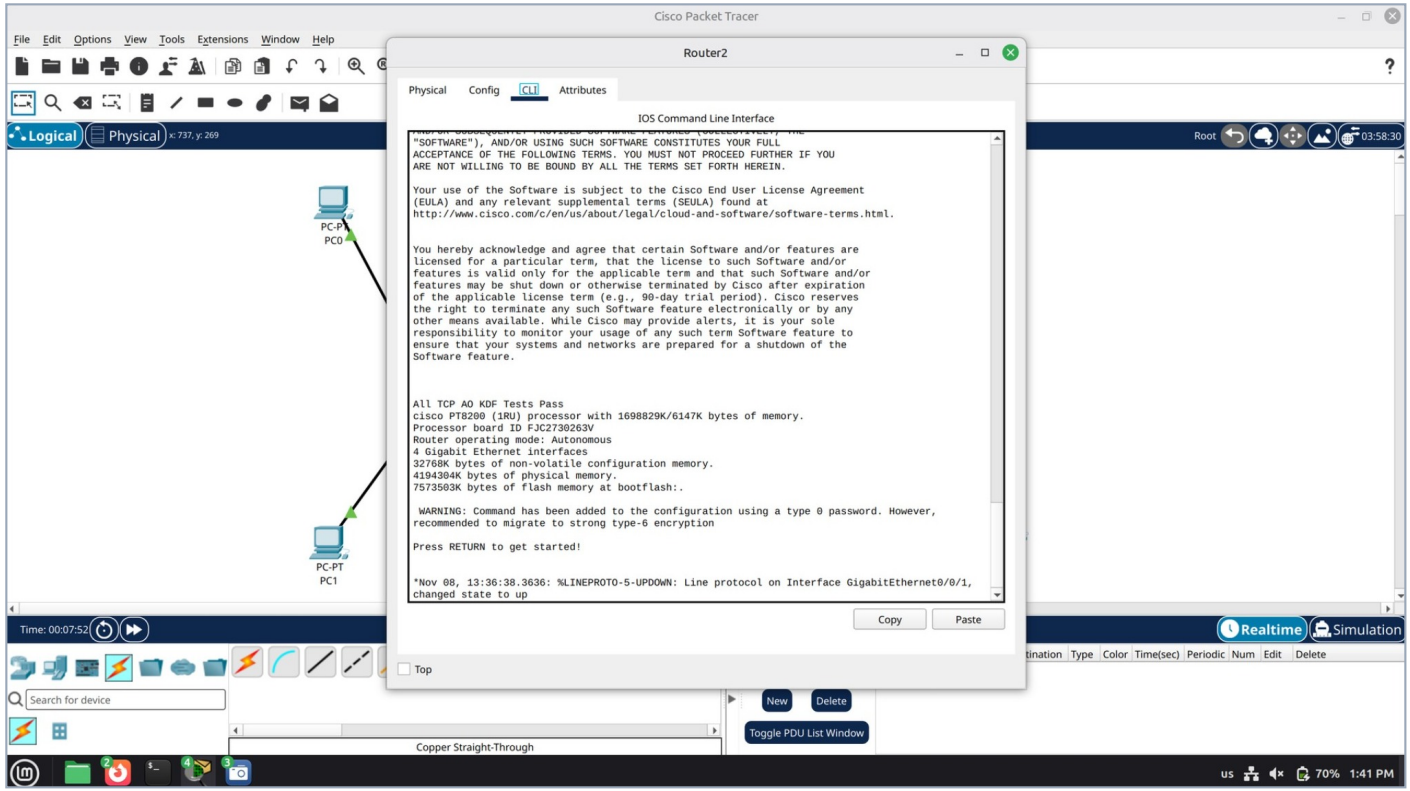
- Open **Switch0 → CLI**
- The boot log reports link and line protocol **up** on Gi0/1, Gi0/2 and Fa0/1
- Model **WS-C2960-24TT-L**, IOS 15.0(2)SE4
- To see which port goes where, run:




```
show interfaces status
```

Open the router CLI

- Open **Router2 → CLI** and press Enter at “Press RETURN to get started!”
- The PT8200 reports **4 Gigabit Ethernet** interfaces
- Names are three-part, **Gi0/0/x**, not G0/0 as in the original guide
- Gi0/0/1 already shows line protocol up

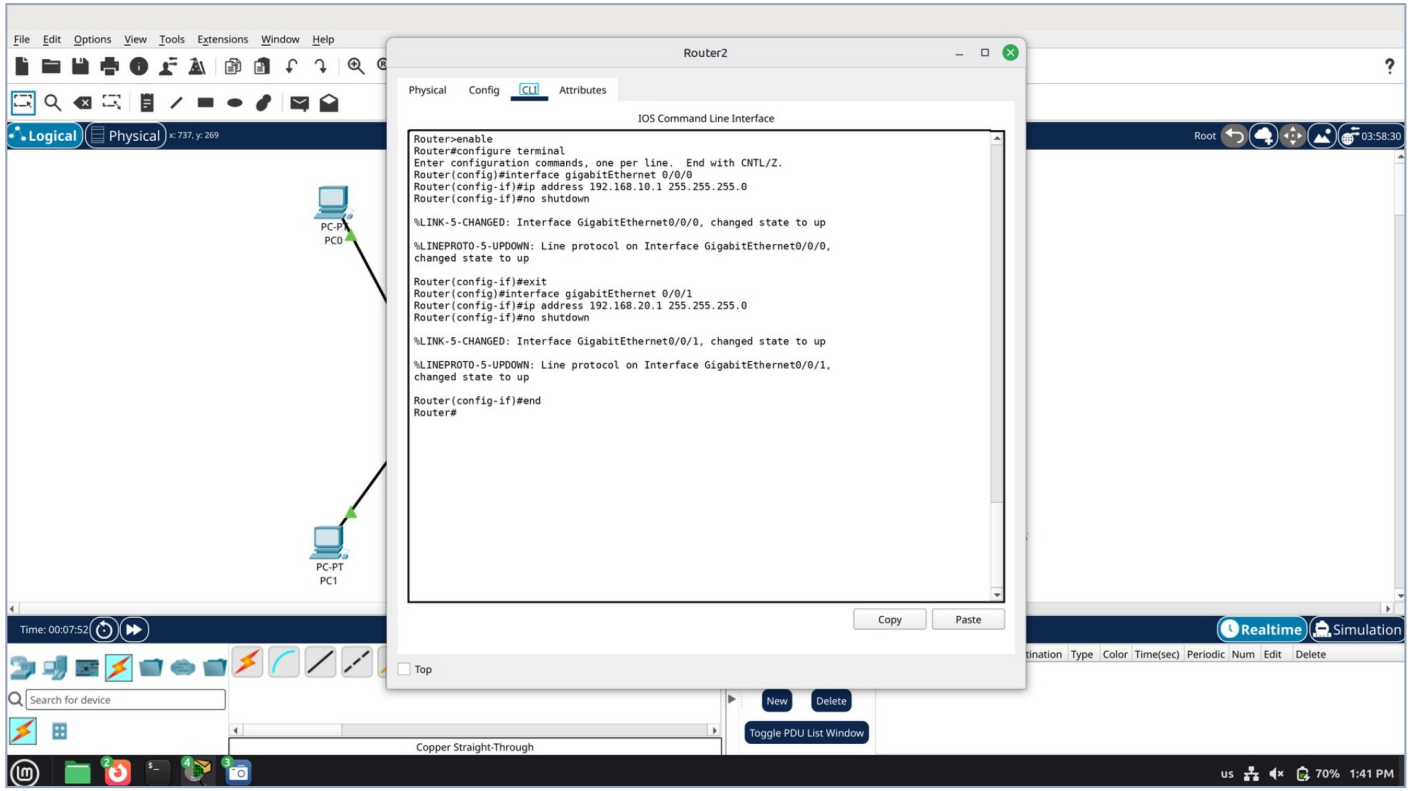



```
enable
```

STEP 3 · ROUTER2

Configure the router

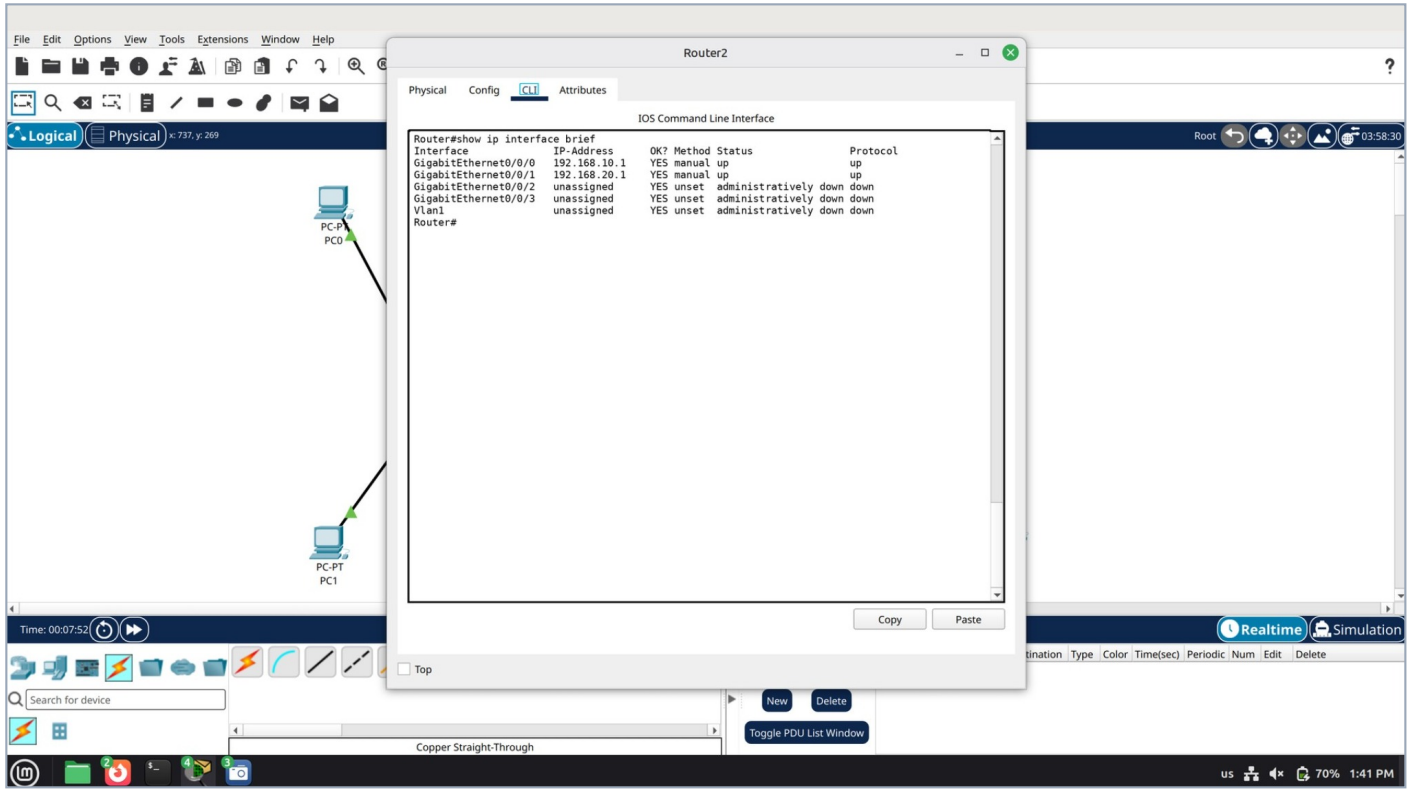
Assumes Gi0/0/0 faces Switch0 and Gi0/0/1 faces Switch1. Swap them if your cables differ.




```
enable
configure terminal
interface gigabitEthernet 0/0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

Verify the router

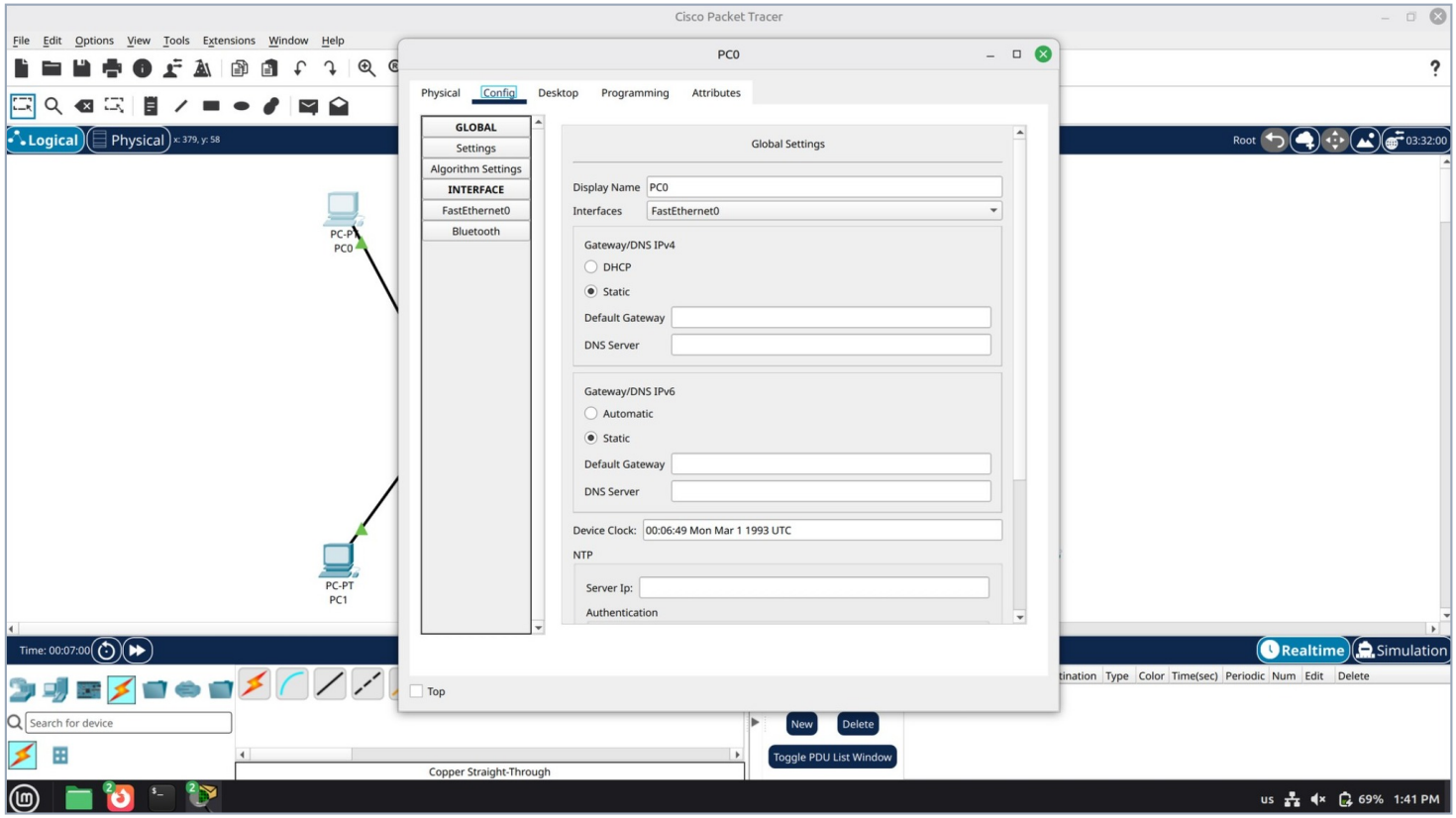
- Both configured ports should read **up / up**
- **administratively down** means the port still needs no shutdown
- **up / down** usually means the cable or the switch side is not ready




```
show ip interface brief
```

Open the PC config

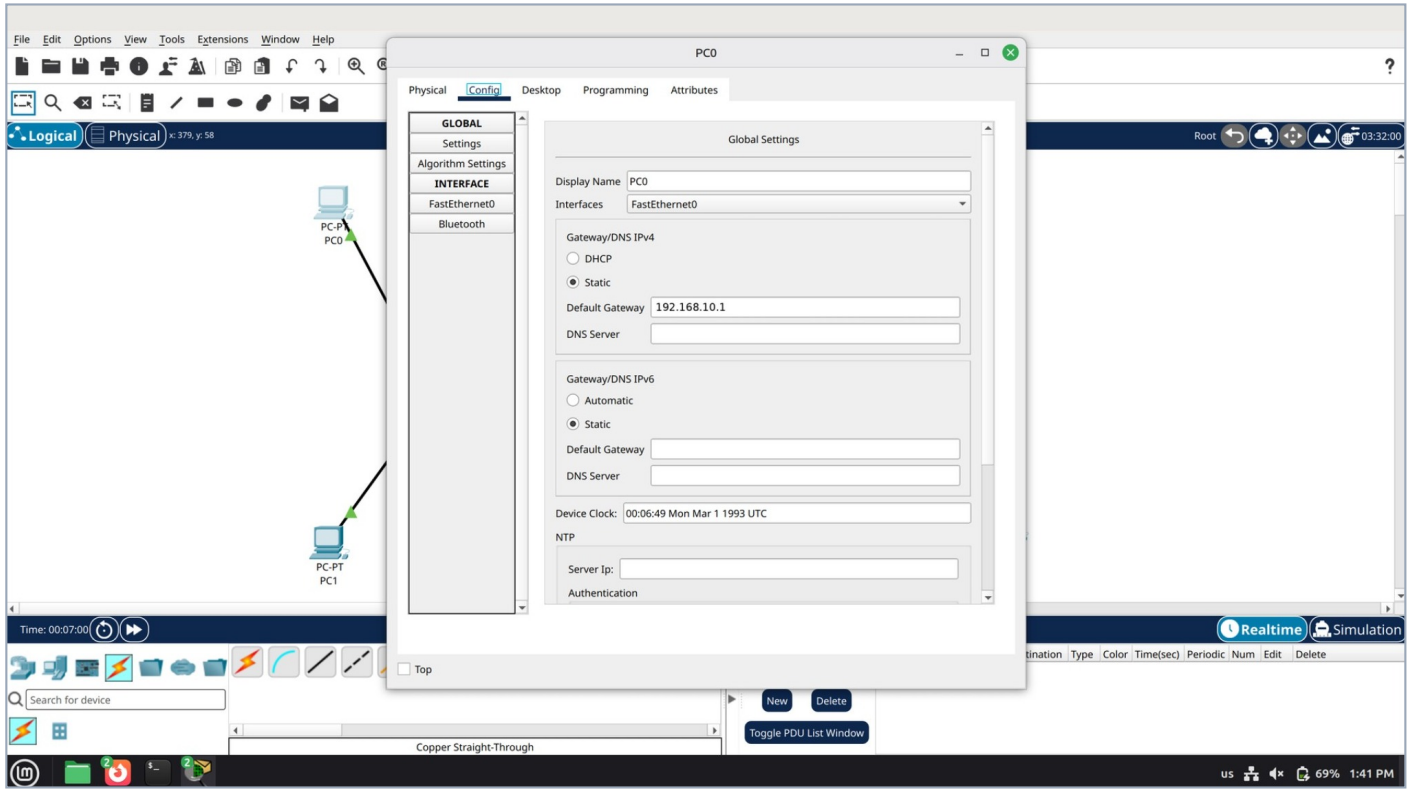
- Open **PC0 → Config → Global Settings**
- IPv4 is **Static**; the gateway and DNS fields are still empty
- The IP address and mask go on the **FastEthernet0** page, or Desktop → IP Configuration
- PC0 gets **192.168.10.10** with mask 255.255.255.0



STEP 5 · ALL FOUR PCS

Set the PC gateway

Mask on every PC: 255.255.255.0. The gateway is Router2’s address on that PC’s LAN.

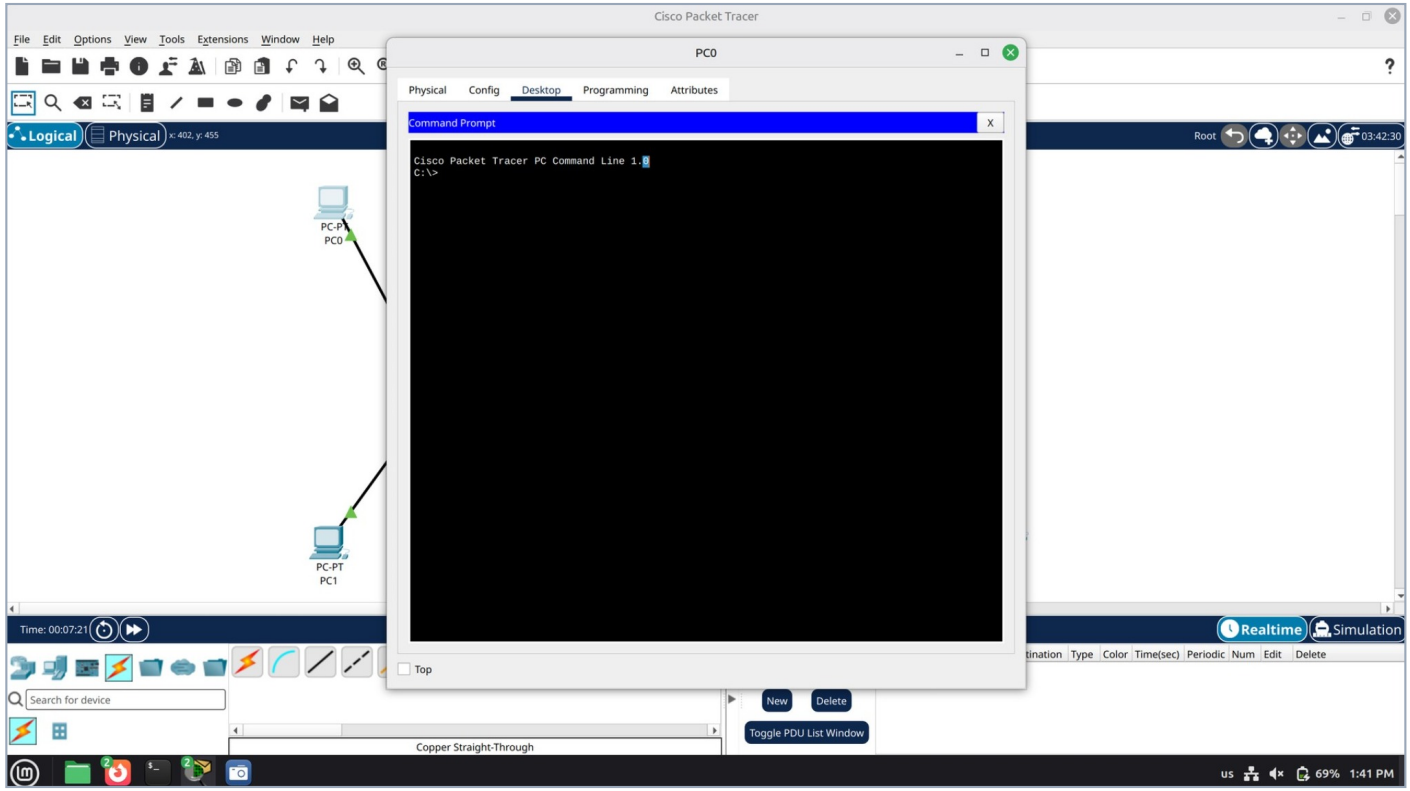


Device	IP address	Gateway
PC0	192.168.10.10	192.168.10.1
PC1	192.168.10.11	192.168.10.1
PC2	192.168.20.10	192.168.20.1
PC3	192.168.20.11	192.168.20.1

STEP 6 · PC0

Open the PC prompt

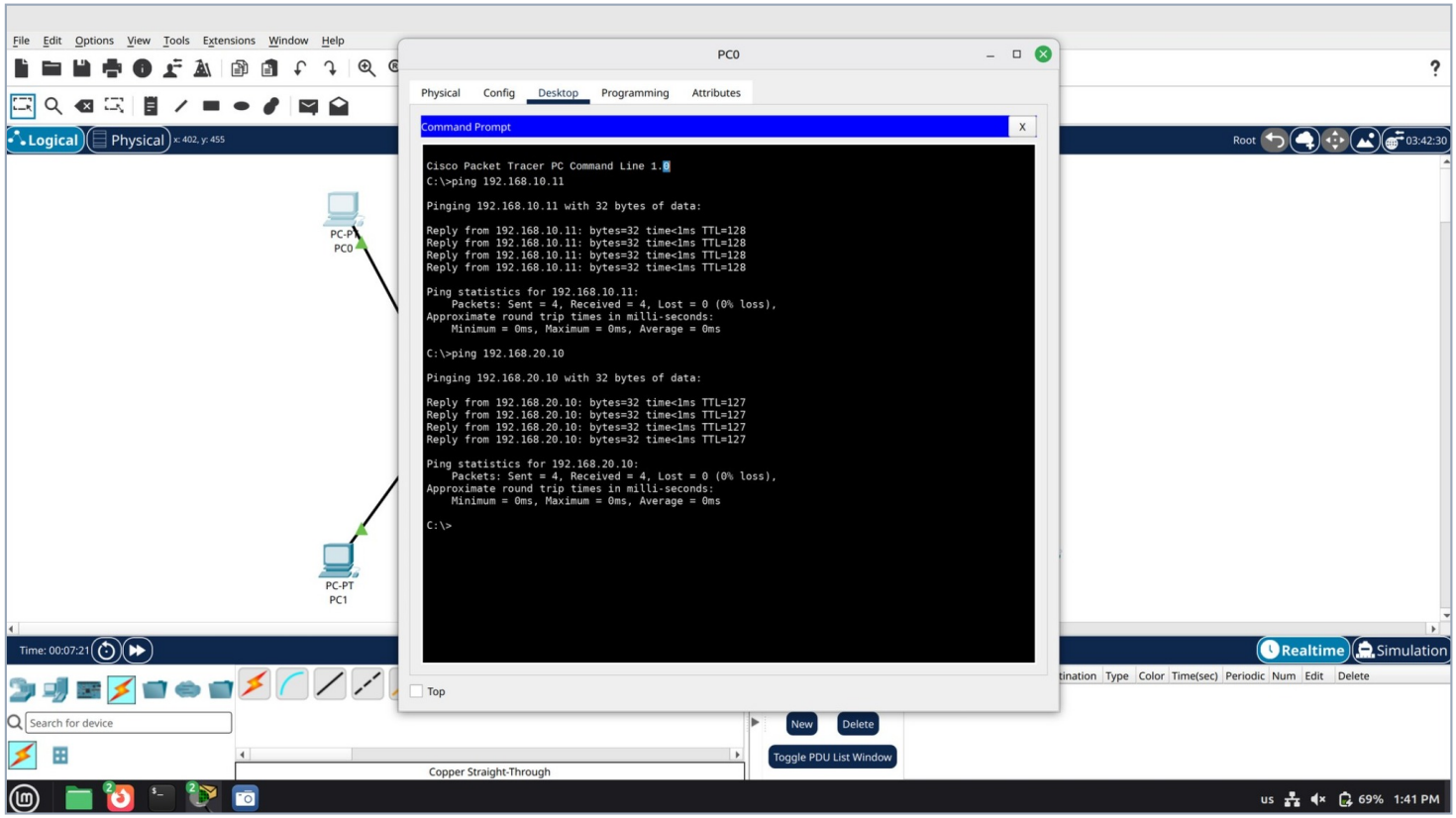
- Open **PC0 → Desktop → Command Prompt**
- The first ping can time out while ARP resolves; run it again




```
ping 192.168.10.11
ping 192.168.10.1
ping 192.168.20.1
ping 192.168.20.10
ping 192.168.20.11
```

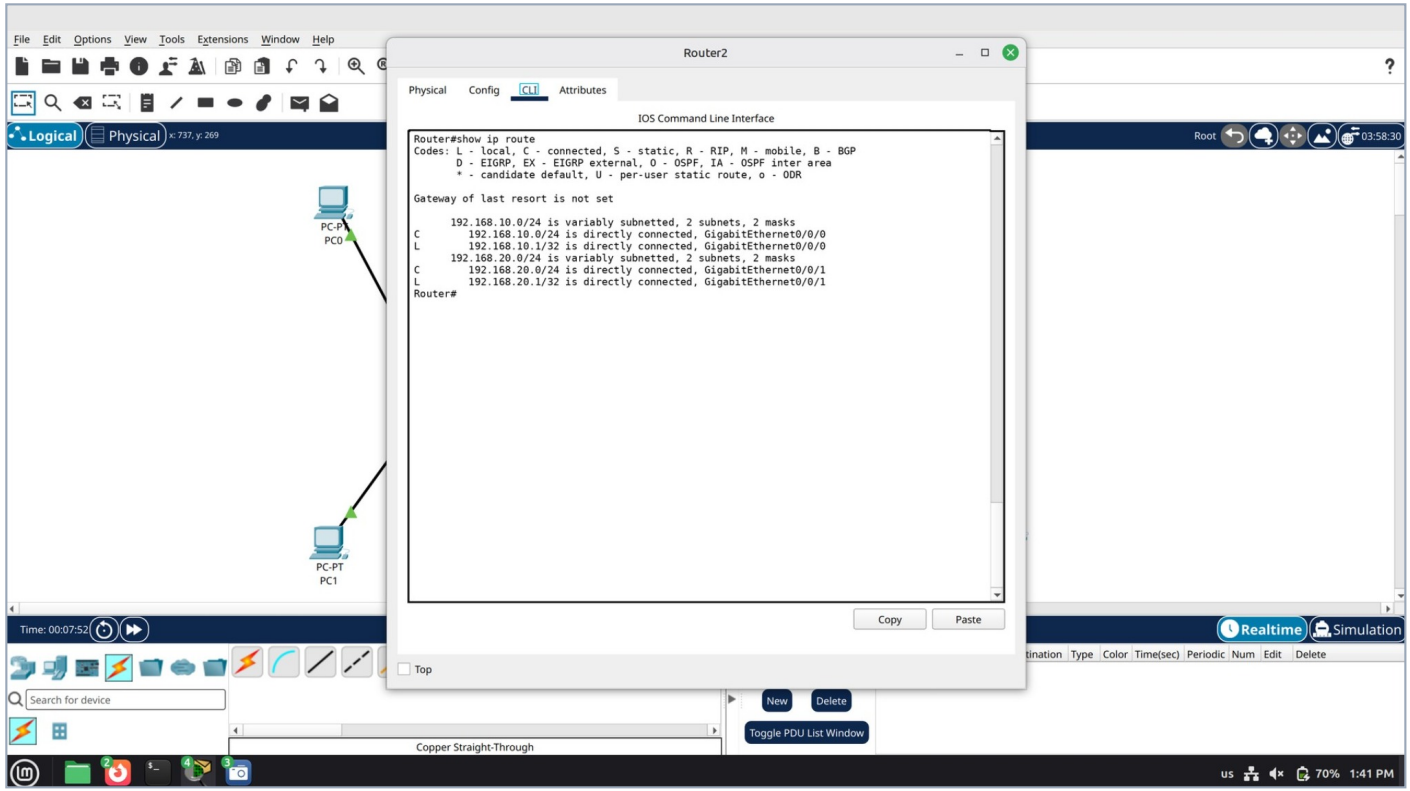
Test connectivity

- **Same LAN:** ping PC1 at 192.168.10.11, TTL 128
- **Across the router:** ping PC2 at 192.168.20.10, TTL 127
- Four replies and **0% loss** means the path works
- TTL drops by 1 because Router2 is one hop away



Why it works

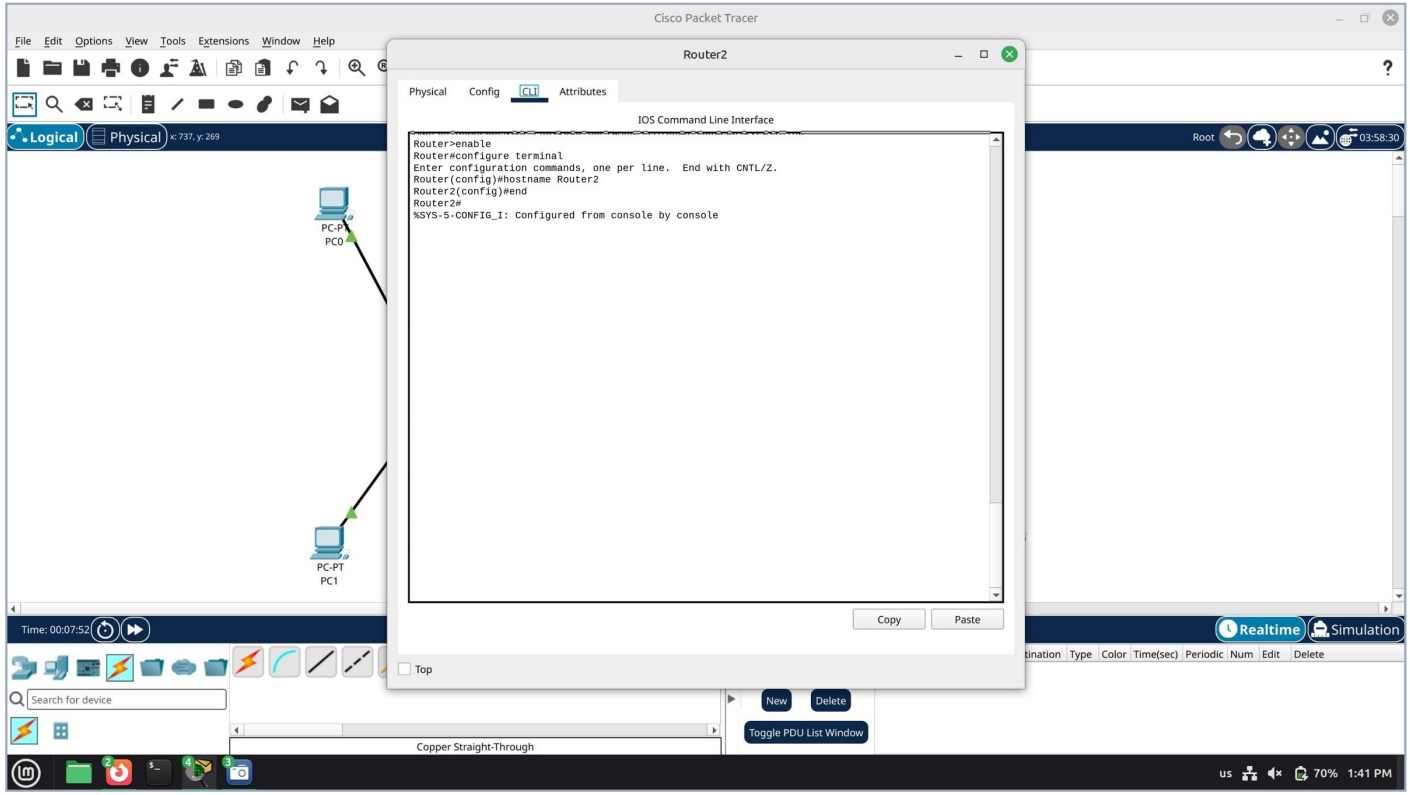
- **Switch0** joins PC0 and PC1 in 192.168.10.0/24
- **Switch1** joins PC2 and PC3 in 192.168.20.0/24
- **Router2** joins the two networks
- No routing protocol needed: both networks are **directly connected**




```
show ip route
```

Name the router

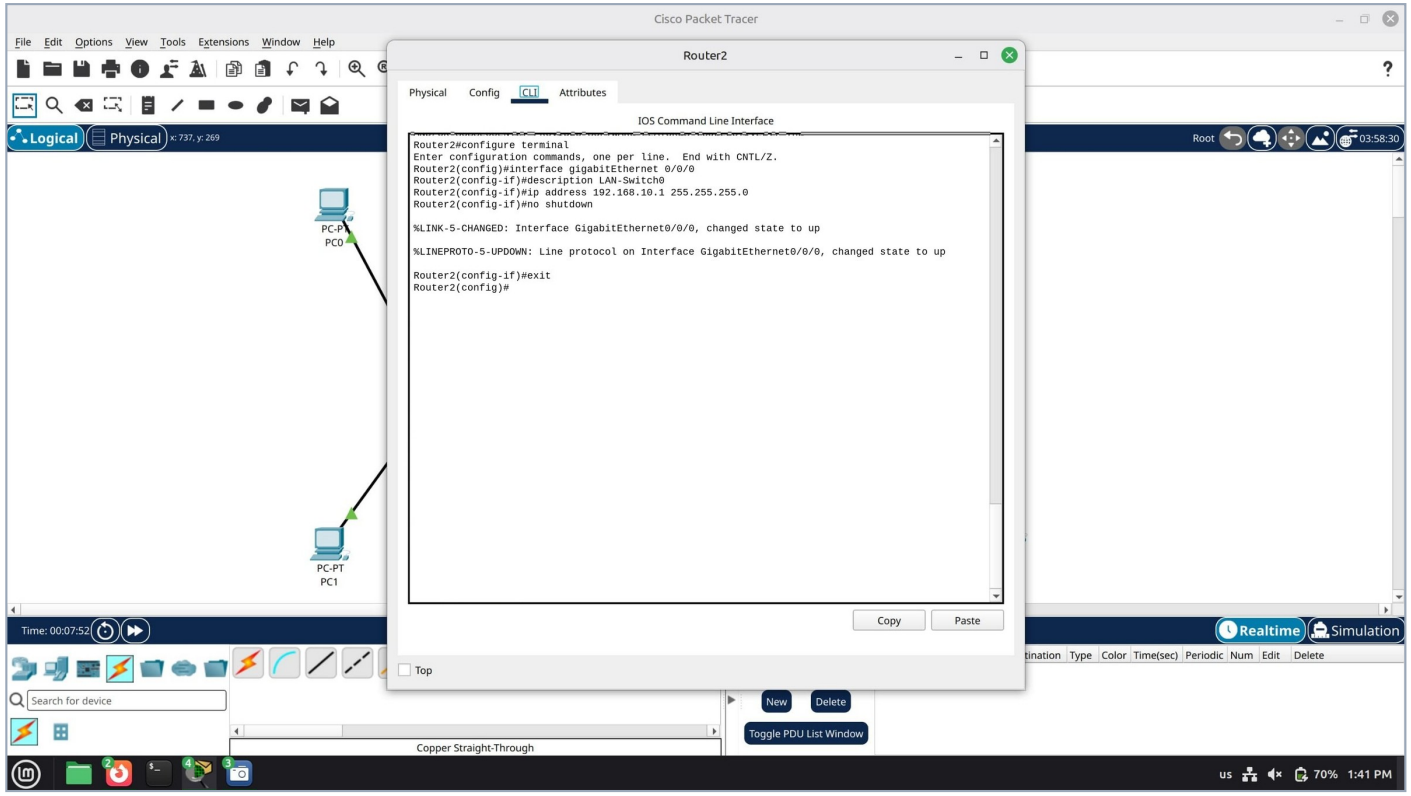
- **Open Router2, CLI tab** and press Enter at the prompt.
- **enable** enters privileged mode; the prompt changes from > to #.
- **configure terminal** enters global configuration mode.
- **hostname** renames the prompt and changes nothing else.




```
enable
configure terminal
hostname Router2
end
```

Configure Gi0/0/0

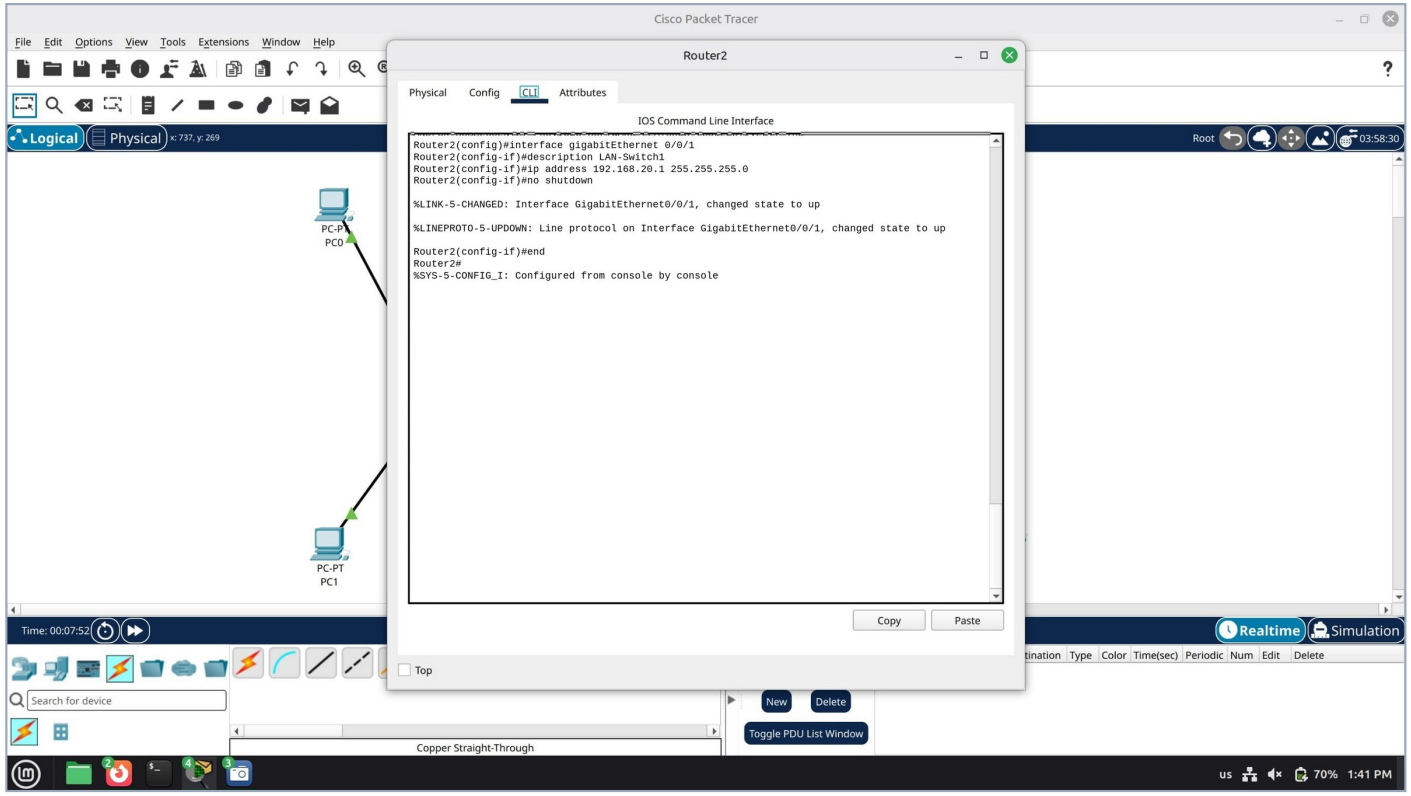
- **Prompt becomes (config-if)** once you select the interface.
- **ip address** sets the gateway of the left LAN (Switch0 side).
- **no shutdown** turns the port on; two messages confirm link and line protocol up.
- **description** is optional, only a label.




```
interface gigabitEthernet 0/0/0
description LAN-Switch0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

Configure Gi0/0/1

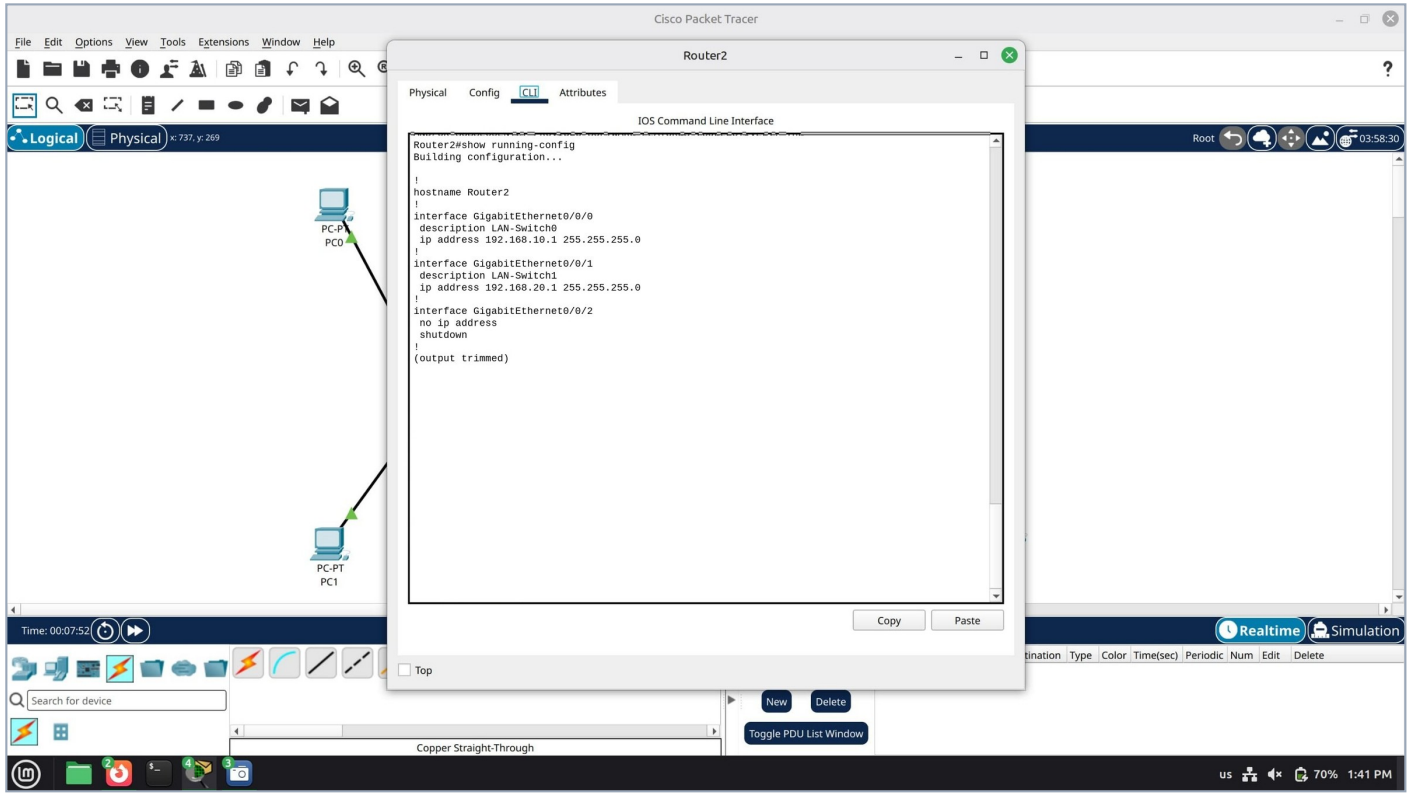
- **Gi0/0/1 faces Switch1**: swap the two ports if your cables differ.
- **192.168.20.1** becomes the gateway for PC2 and PC3.
- **end** leaves configuration mode; the SYS-5-CONFIG_I message is normal.




```
interface gigabitEthernet 0/0/1
description LAN-Switch1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

Review the config

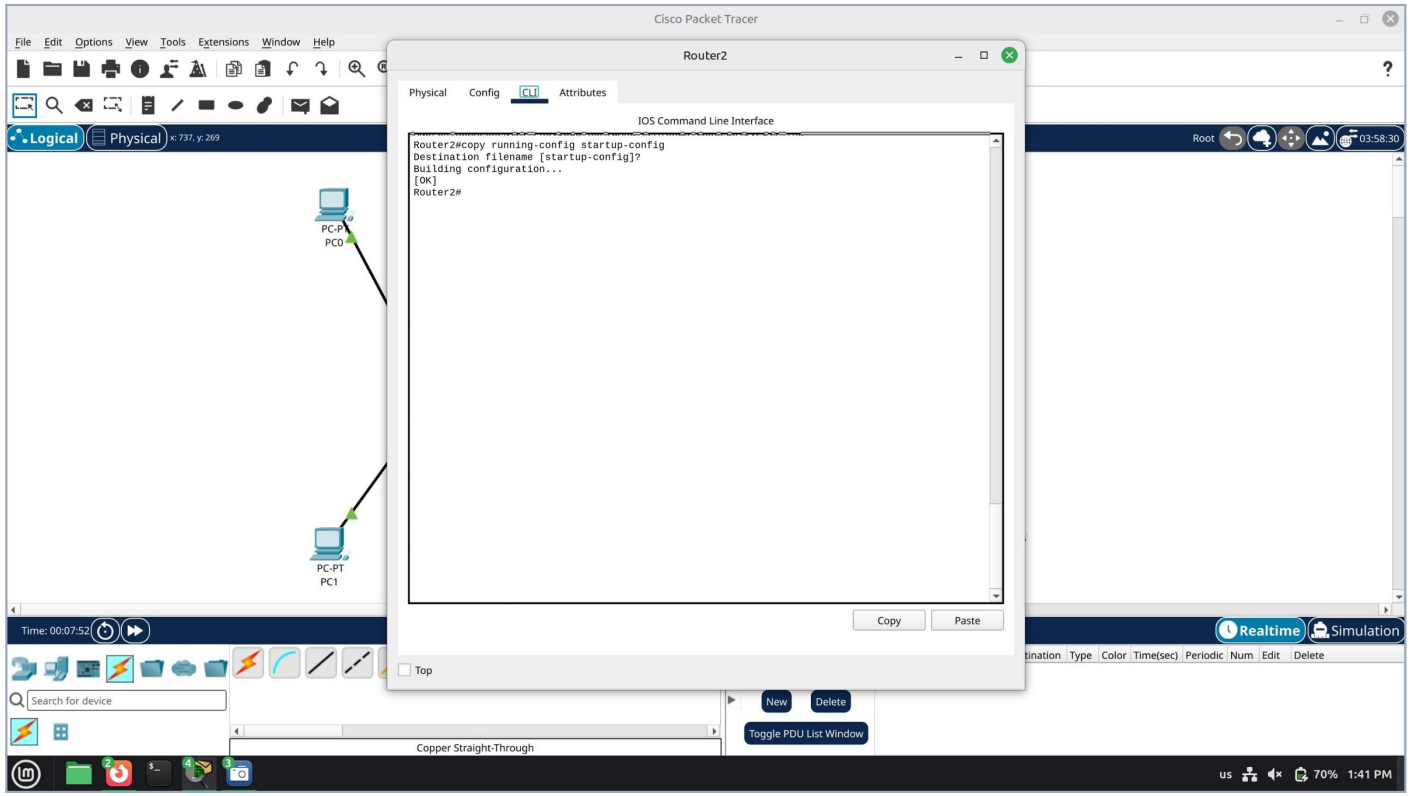
- **show running-config** prints the active configuration held in RAM.
- **Check both interface blocks**: address, mask and no shutdown.
- **Unused ports** show no ip address and shutdown.




```
show running-config
```

Save the configuration

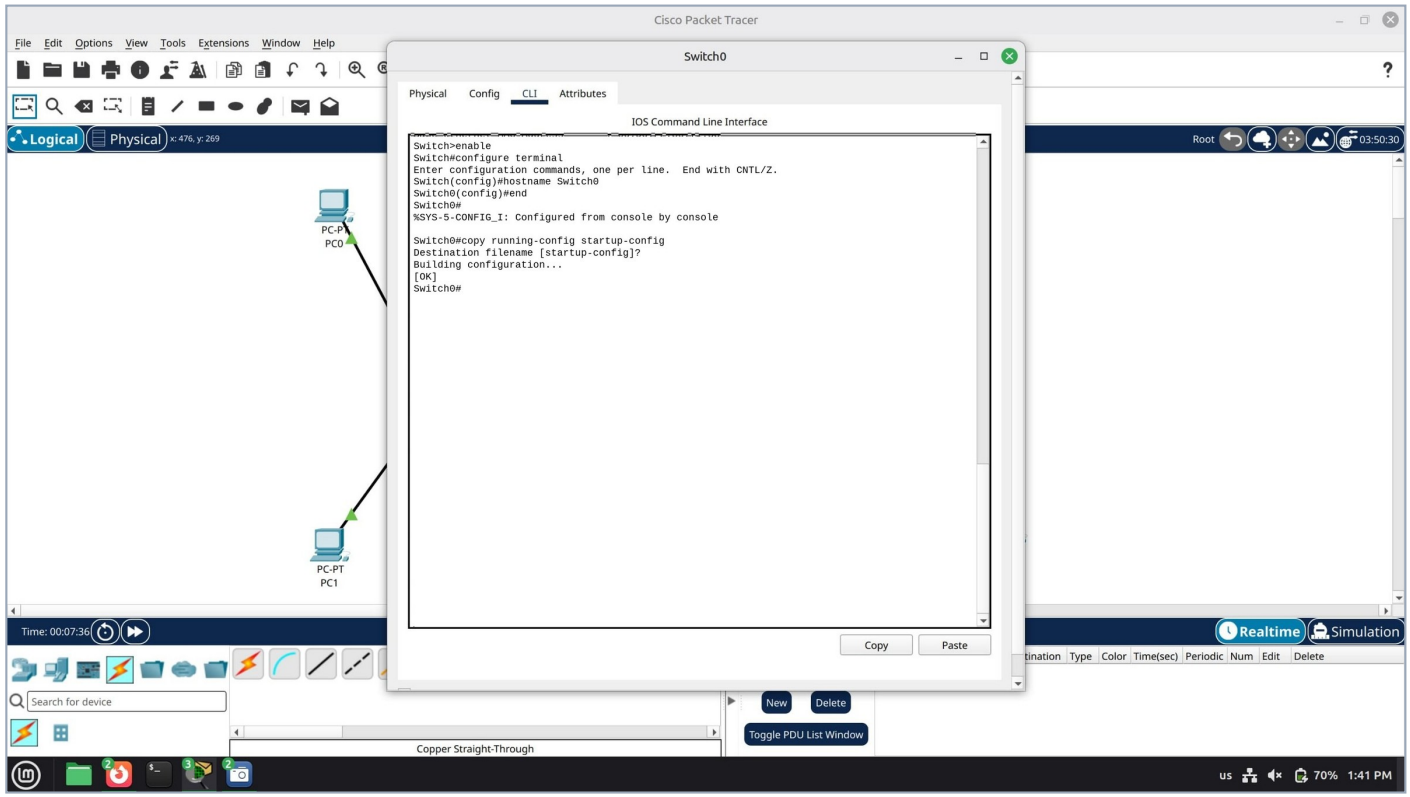
- **running-config** lives in RAM and is lost on reload.
- **startup-config** lives in NVRAM and survives a reload.
- **Press Enter** to accept the filename; [OK] confirms the save.




```
copy running-config startup-config
```

Name and save Switch0

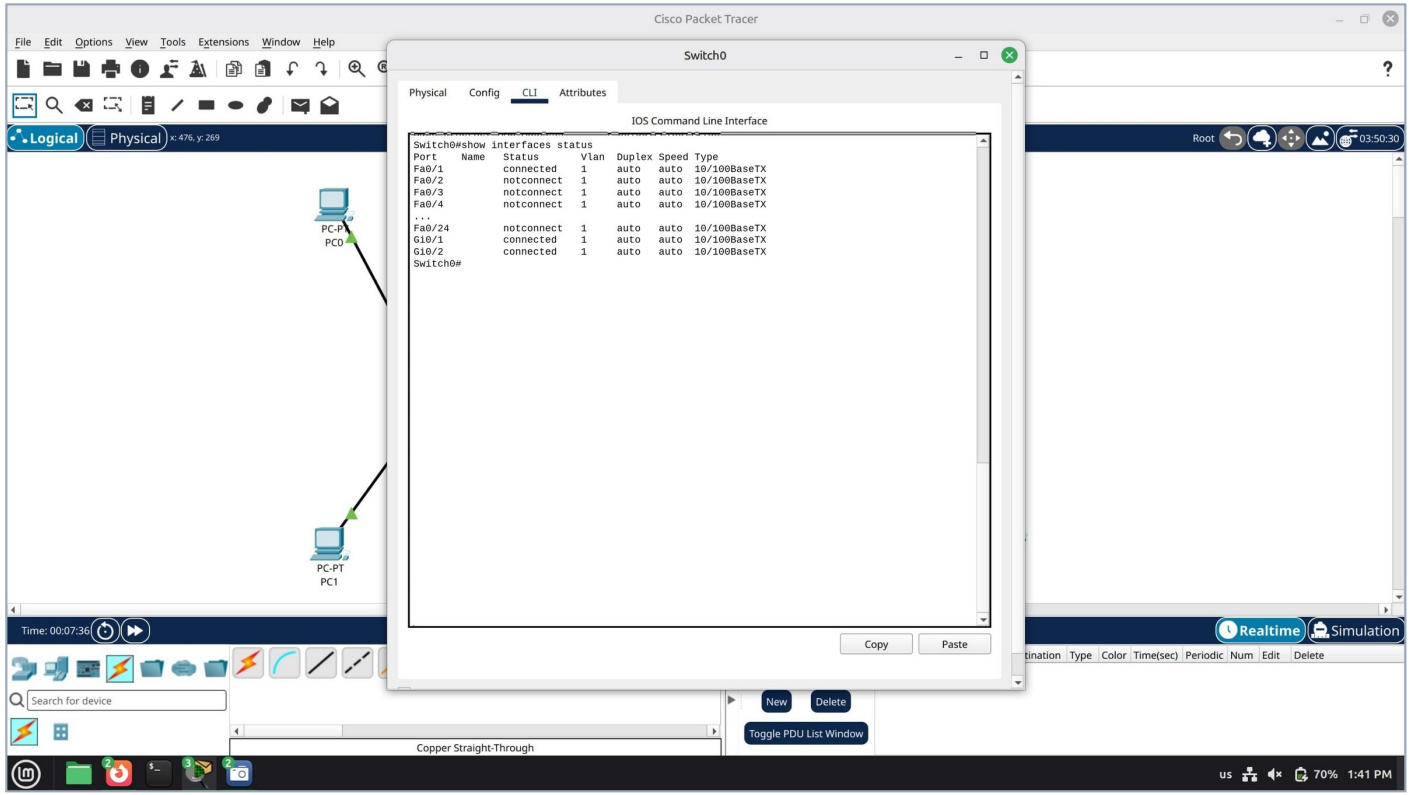
- **No IP or routing setup** is needed: a switch forwards frames by default.
- **hostname** makes each switch easy to tell apart.
- **Repeat on Switch1** with hostname Switch1.




```
enable
configure terminal
hostname Switch0
end
copy running-config startup-config
```

Check port status

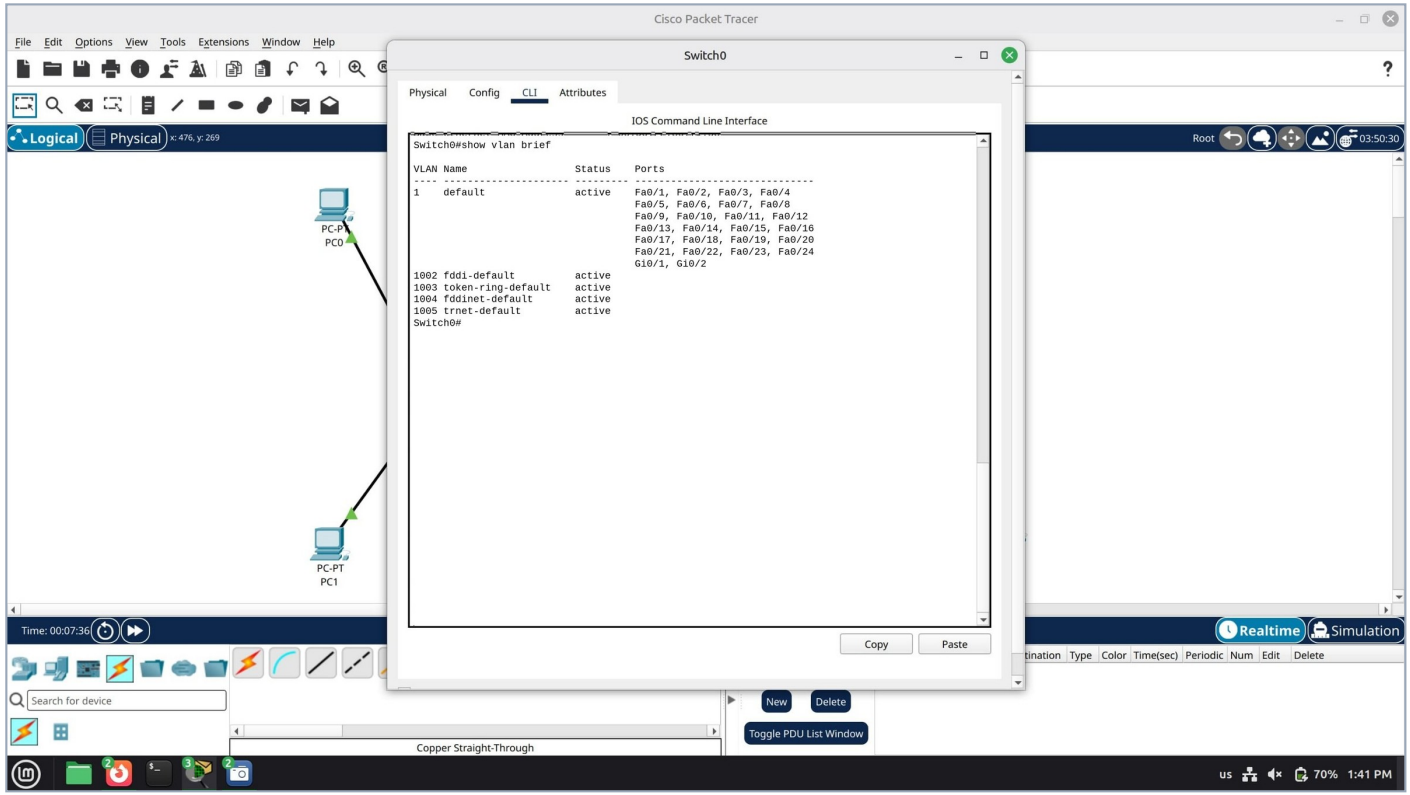
- **connected** means the cable is in and the other end is up.
- **notconnect** means nothing is plugged into that port.
- **Three ports are connected**: PC0, PC1 and Router2.
- **Vlan 1** on every port is the default.




```
show interfaces status
```

Check the VLAN

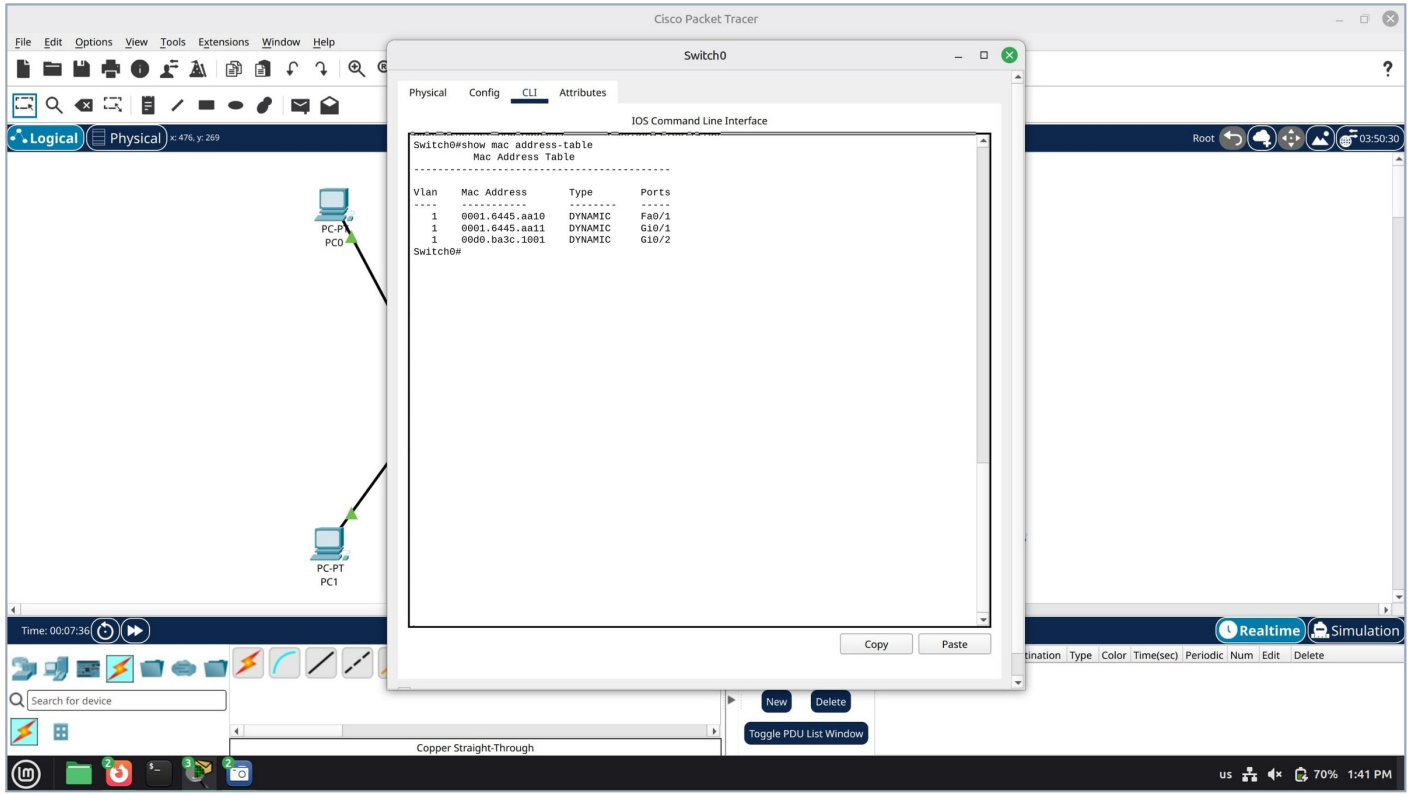
- **VLAN 1 holds every port** by default.
- **One VLAN = one broadcast domain**, so one subnet per switch.
- **That is why a router is needed** to join Switch0 and Switch1.
- **1002 to 1005** are legacy default VLANs.




```
show vlan brief
```

Read the MAC table

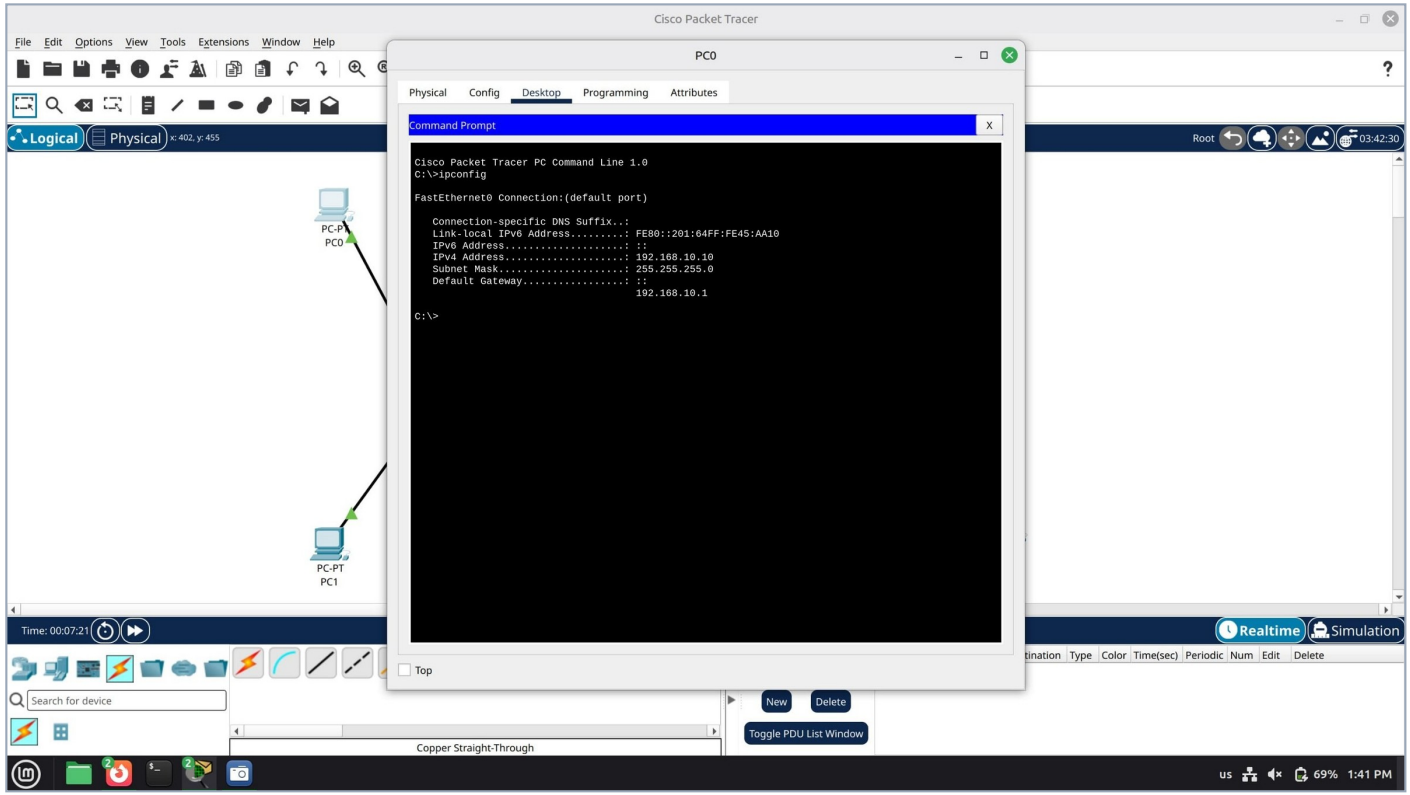
- **Run it after the pings.**
- **The switch learns** each source MAC and the port it arrived on.
- **Router2's MAC stands in** for everything on the other LAN.
- **PC2 and PC3 never appear** here: the router rewrites the frame.




```
show mac address-table
```

Check the PC address

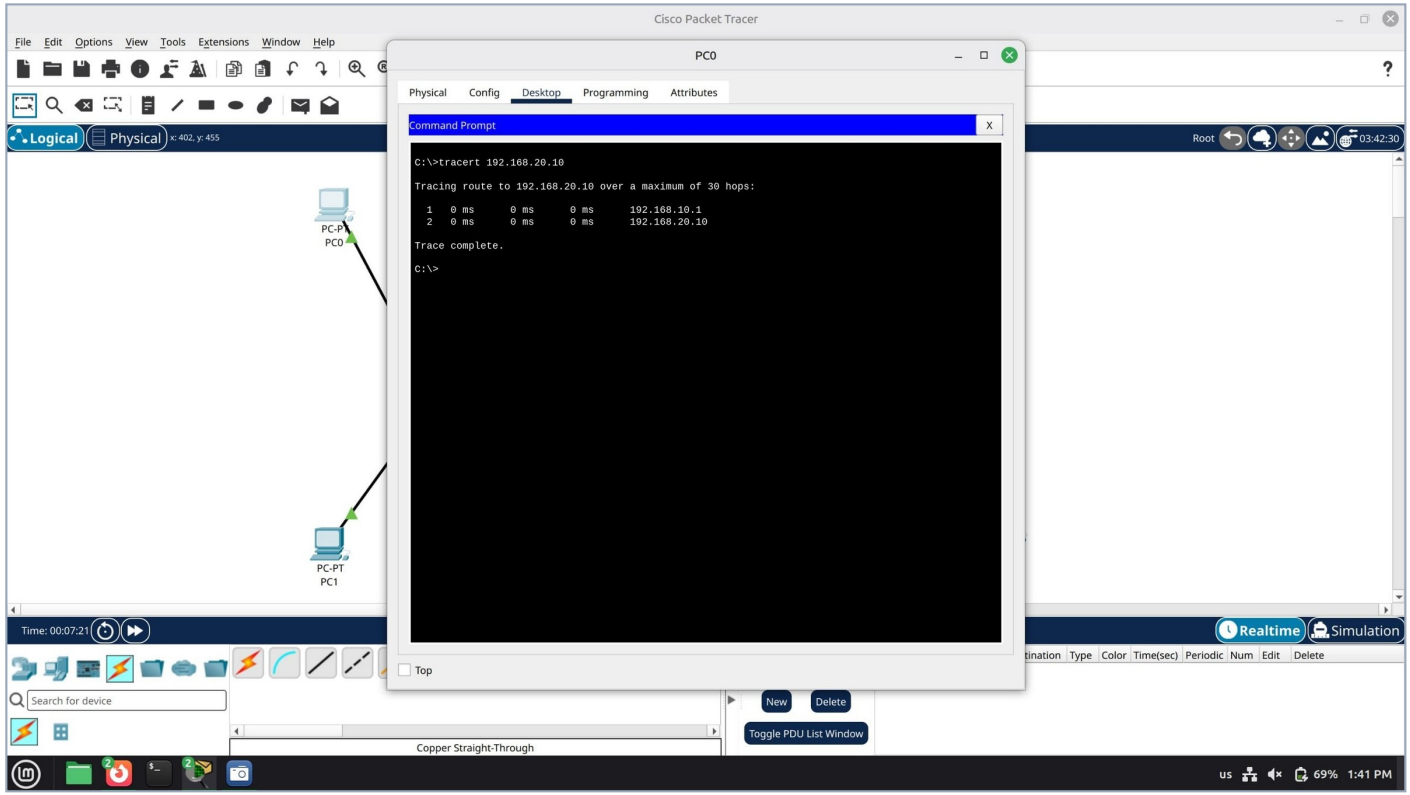
- **Open PC0, Desktop, Command Prompt**.
- **ipconfig** shows what the PC is really using.
- **IPv4, mask and gateway** must match the addressing plan.
- **Empty gateway** means pings to the other LAN will fail.




```
ipconfig
```

Trace the path

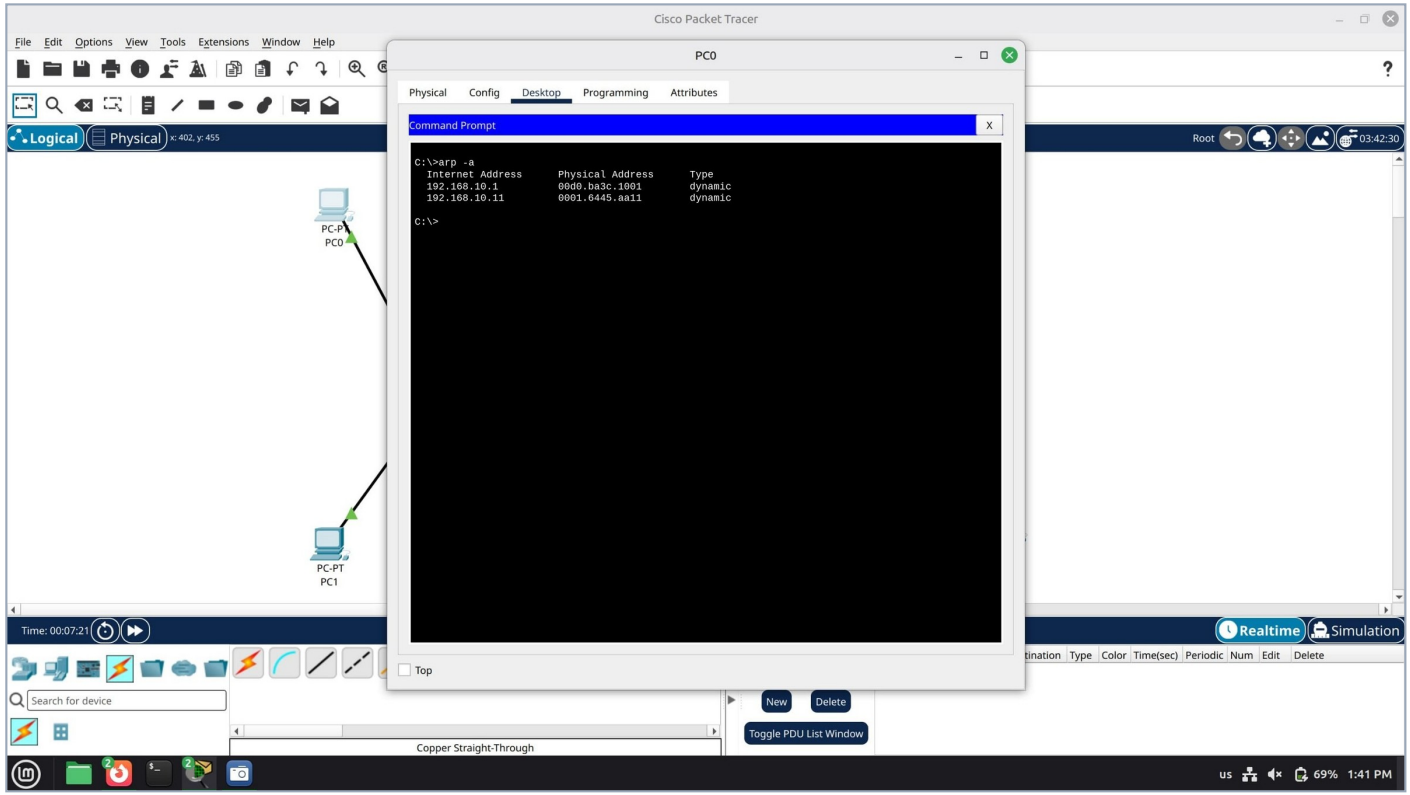
- **tracert** lists every router between PC0 and PC2.
- **Hop 1 is 192.168.10.1**, the gateway on Router2.
- **Hop 2 is the target**: one router, so TTL drops by 1.




```
tracert 192.168.20.10
```

Read the ARP table

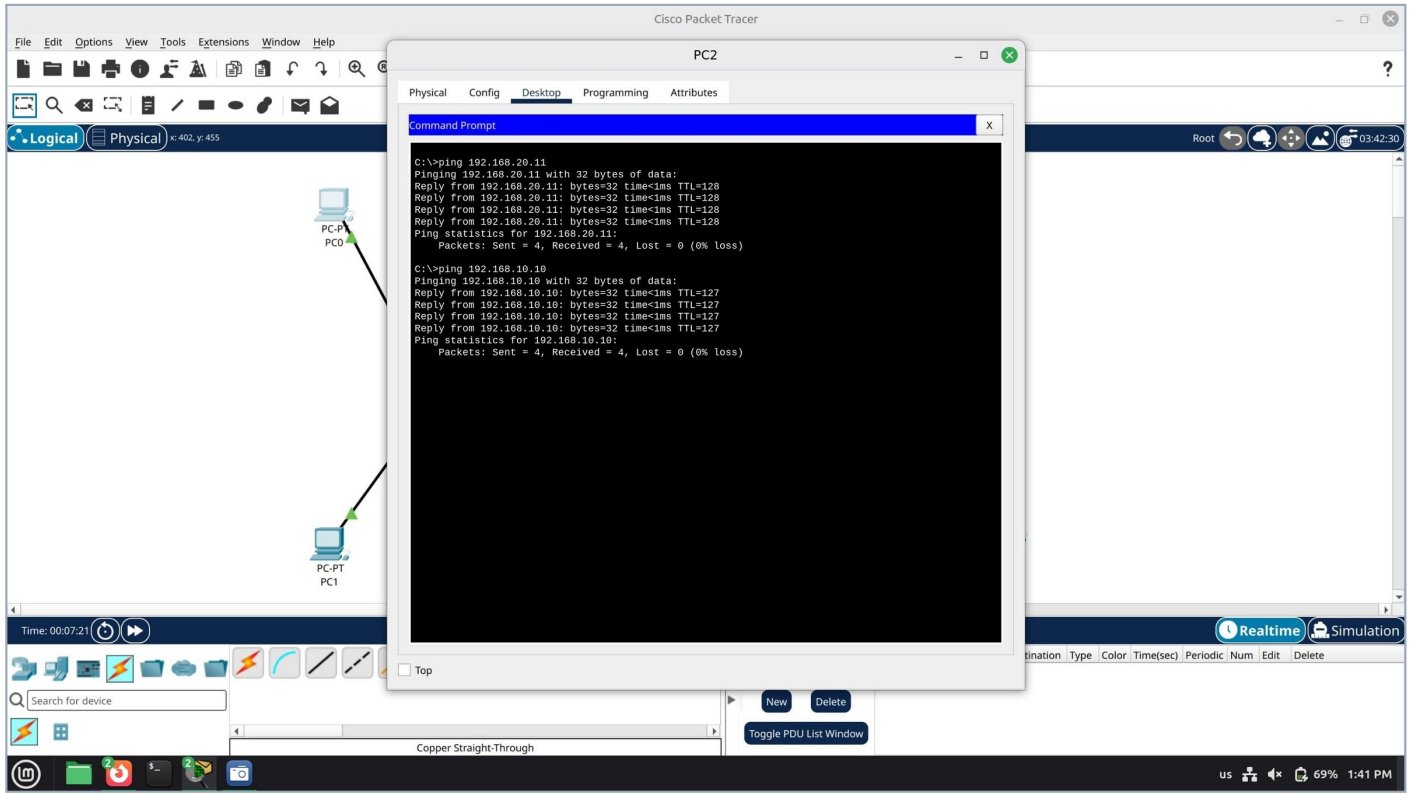
- **arp -a** lists IP to MAC mappings learned from pings.
- **192.168.10.1** is Router2, the gateway PC0 uses for the other LAN.
- **No 192.168.20.x entry**: PC0 never ARPs for hosts on another LAN.




```
arp -a
```

Ping back from PC2

- **Same LAN**: PC3 replies with TTL 128.
- **Other LAN**: PC0 replies with TTL 127 (one hop).
- **Reverse direction works** with no extra router setup.




```
ping 192.168.20.11
ping 192.168.10.10
```